

RELATIONS AND FUNCTIONS

1.	A function $f: A \rightarrow B$ defined as $f(x) = 2x$ is both one-one and onto. If $A = \{1, 2, 3, 4\}$, then find the set B.
2.	A relation R is defined on a set of real numbers $\mathbb{R}$ as $R = \{(x, y) : x \cdot y \text{ is an irrational number}\}.$ Check whether R is reflexive, symmetric and transitive or not.
3.	A function $f: [-4, 4] \rightarrow [0, 4]$ is given by $f(x) = \sqrt{16 - x^2}$. Show that f is an onto function but not a one-one function. Further, find all possible values of 'a' for which $f(a) = \sqrt{7}$.
4.	Check whether a function $f: \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as $f(x) = \frac{x}{1+x^2}$ is one-one and onto or not.
5.	If N denotes the set of all natural numbers and R is the relation on $N \times N$ defined by $(a, b) R (c, d)$, if $ad(b + c) = bc(a + d)$. Show that R is an equivalence relation.
6.	Let $f: \mathbb{R} - \left\{-\frac{4}{3}\right\} \rightarrow \mathbb{R}$ be a function defined as $f(x) = \frac{4x}{3x+4}$. Show that f is a one-one function. Also, check whether f is an onto function or not.
7.	Let $A = \{3, 5\}$. Then number of reflexive relations on A is (a) 2 (b) 4 (c) 0 (d) 8
8.	Check whether the relation S in the set of all real numbers ($\mathbb{R}$) defined by $S = \{(a, b) : a \leq b^3\}$ is reflexive, symmetric or transitive.
9.	Prove that the greatest integer function $f: \mathbb{R} \rightarrow \mathbb{R}$, given by $f(x) = [x]$, is neither one-one nor onto.
10.	Show that the relation S in set $\mathbb{R}$ of real numbers defined by $S = \{(a, b) : a \leq b^3, a \in \mathbb{R}, b \in \mathbb{R}\}$ is neither reflexive, nor symmetric, nor transitive.

11.	Let R be the relation defined in the set $A = \{1, 2, 3, 4, 5, 6, 7\}$ by $R = \{(a, b) : \text{both } a \text{ and } b \text{ are either odd or even}\}$. Show that R is an equivalence relation. Hence, find the elements of equivalence class [1].
12.	A function $f : R_+ \rightarrow R$ (where R_+ is the set of all non-negative real numbers) defined by $f(x) = 4x + 3$ is : (A) one-one but not onto (B) onto but not one-one (C) both one-one and onto (D) neither one-one nor onto
13.	A relation R on set $A = \{1, 2, 3, 4, 5\}$ is defined as $R = \{(x, y) : x^2 - y^2 < 8\}$. Check whether the relation R is reflexive, symmetric and transitive.
14.	A function f is defined from $R \rightarrow R$ as $f(x) = ax + b$, such that $f(1) = 1$ and $f(2) = 3$. Find function $f(x)$. Hence, check whether function $f(x)$ is one-one and onto or not.
15.	Which of the following statements is not true about equivalence classes A_i ($i = 1, 2, \dots, n$) formed by an equivalence relation R defined on a set A ? (A) $\bigcup_{i=1}^n A_i = A$ (B) $A_i \cap A_j \neq \phi, i \neq j$ (C) $x \in A_i \text{ and } x \in A_j \Rightarrow A_i = A_j$ (D) All elements of A_i are related to each other, for all i
16.	Show that a function $f : R \rightarrow R$ defined by $f(x) = \frac{2x}{1+x^2}$ is neither one-one nor onto. Further, find set A so that the given function $f : R \rightarrow A$ becomes an onto function.
17.	A relation R is defined on $N \times N$ (where N is the set of natural numbers) as : $(a, b) R (c, d) \Leftrightarrow a - c = b - d$ Show that R is an equivalence relation.

18.	Let Z denote the set of integers, then function $f : Z \rightarrow Z$ defined as $f(x) = x^3 - 1$ is : (A) both one-one and onto (B) one-one but not onto (C) onto but not one-one (D) neither one-one nor onto
19.	Let $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{a\}$. Find the value of 'a' such that the function $f : A \rightarrow B$ defined by $f(x) = \frac{x-2}{x-3}$ is onto. Also, check whether the given function is one-one or not.
20.	Assertion (A) : The relation $R = \{(x, y) : (x + y) \text{ is a prime number and } x, y \in \mathbb{N}\}$ is not a reflexive relation. Reason (R) : The number '2n' is composite for all natural numbers n.
21.	Let $A = \mathbb{R} - \{5\}$ and $B = \mathbb{R} - \{1\}$. Consider the function $f : A \rightarrow B$, defined by $f(x) = \frac{x-3}{x-5}$. Show that f is one-one and onto.
22.	Check whether the relation S in the set of real numbers $\mathbb{R}$ defined by $S = \{(a, b) : \text{where } a - b + \sqrt{2} \text{ is an irrational number}\}$ is reflexive, symmetric or transitive.
23.	A function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^2 - 4x + 5$ is : (A) injective but not surjective. (B) surjective but not injective. (C) both injective and surjective. (D) neither injective nor surjective.
24.	Assertion (A) : Let Z be the set of integers. A function $f : Z \rightarrow Z$ defined as $f(x) = 3x - 5, \forall x \in Z$ is a bijective. Reason (R) : A function is a bijective if it is both surjective and injective.
25.	Assertion (A) : Let Z be the set of integers. A function $f : Z \rightarrow Z$ defined as $f(x) = 3x - 5, \forall x \in Z$ is a bijective. Reason (R) : A function is a bijective if it is both surjective and injective.
26.	Prove that $f : \mathbb{N} \rightarrow \mathbb{N}$ defined as $f(x) = ax + b$ ($a, b \in \mathbb{N}$) is one-one but not onto.

34.	Show that the function $f : \mathbb{N} \rightarrow \mathbb{N}$, where $\mathbb{N}$ is a set of natural numbers, given by $f(n) = \begin{cases} n-1, & \text{if } n \text{ is even} \\ n+1, & \text{if } n \text{ is odd} \end{cases}$ is a bijection.
35.	Let $f : A \rightarrow B$ be defined by $f(x) = \frac{x-2}{x-3}$, where $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{1\}$. Discuss the bijectivity of the function.
36.	Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 4x^3 - 5$, $\forall x \in \mathbb{R}$ is one-one and onto.
37.	Let R be a relation defined on a set $\mathbb{N}$ of natural numbers such that $R = \{(x, y) : xy \text{ is a square of a natural number, } x, y \in \mathbb{N}\}$. Determine if the relation R is an equivalence relation.
	CASE STUDY

38.

- (a) Students of a school are taken to a railway museum to learn about railways heritage and its history.



An exhibit in the museum depicted many rail lines on the track near the railway station. Let L be the set of all rail lines on the railway track and R be the relation on L defined by

$$R = \{(l_1, l_2) : l_1 \text{ is parallel to } l_2\}$$

On the basis of the above information, answer the following questions :

- (i) Find whether the relation R is symmetric or not.
- (ii) Find whether the relation R is transitive or not.
- (iii) If one of the rail lines on the railway track is represented by the equation $y = 3x + 2$, then find the set of rail lines in R related to it.

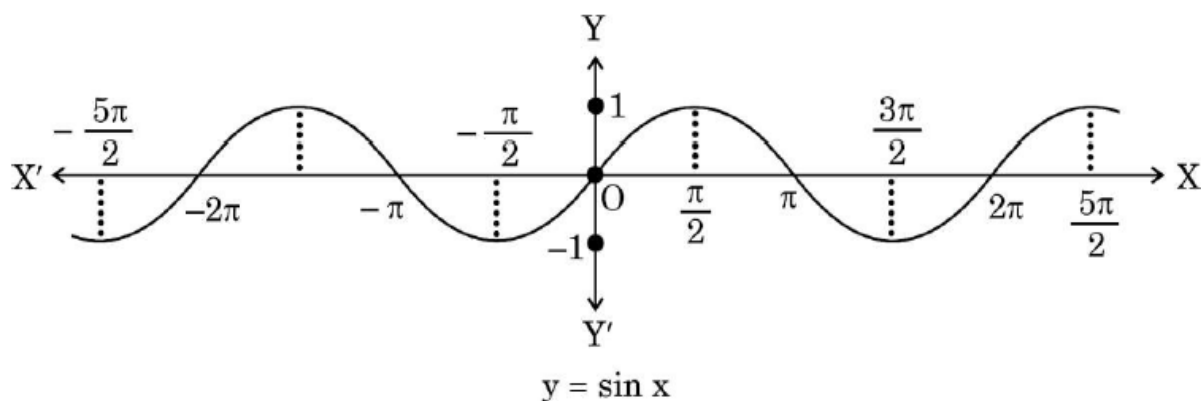
OR

- (b) Let S be the relation defined by $S = \{(l_1, l_2) : l_1 \text{ is perpendicular to } l_2\}$ check whether the relation S is symmetric and transitive.

39.

If a function $f : X \rightarrow Y$ defined as $f(x) = y$ is one-one and onto, then we can define a unique function $g : Y \rightarrow X$ such that $g(y) = x$, where $x \in X$ and $y = f(x)$, $y \in Y$. Function g is called the inverse of function f .

The domain of sine function is $\mathbb{R}$ and function $\sin : \mathbb{R} \rightarrow \mathbb{R}$ is neither one-one nor onto. The following graph shows the sine function.



Let sine function be defined from set A to $[-1, 1]$ such that inverse of sine function exists, i.e., $\sin^{-1} x$ is defined from $[-1, 1]$ to A.

On the basis of the above information, answer the following questions :

- (i) If A is the interval other than principal value branch, give an example of one such interval.
- (ii) If $\sin^{-1}(x)$ is defined from $[-1, 1]$ to its principal value branch, find the value of $\sin^{-1}\left(-\frac{1}{2}\right) - \sin^{-1}(1)$.
- (iii) (a) Draw the graph of $\sin^{-1} x$ from $[-1, 1]$ to its principal value branch.

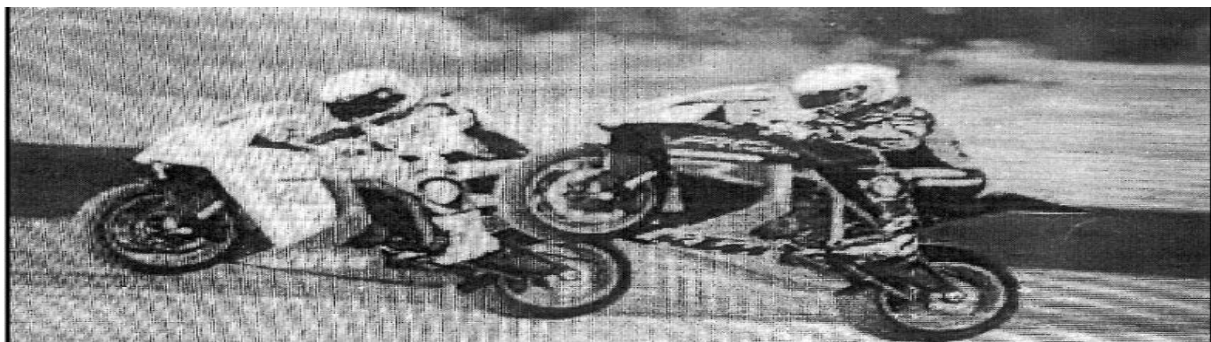
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- (iii) (b) Find the domain and range of $f(x) = 2 \sin^{-1}(1 - x)$.

40.

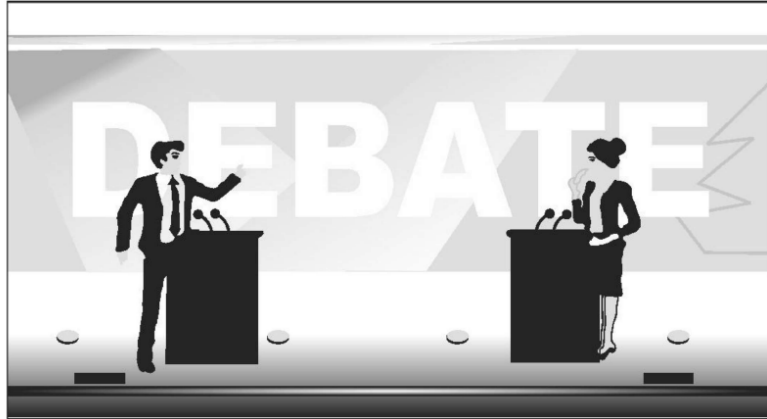
An organization conducted bike race under two different categories – Boys and Girls. There were 28 participants in all. Among all of them, finally three from category 1 and two from category 2 were selected for the final race. Ravi forms two sets B and G with these participants for his college project.

Let $B = \{b_1, b_2, b_3\}$ and $G = \{g_1, g_2\}$, where B represents the set of Boys selected and G the set of Girls selected for the final race.



	Based on the above information, answer the following questions : (I) How many relations are possible from B to G ? (II) Among all the possible relations from B to G, how many functions can be formed from B to G ? (III) Let $R : B \rightarrow B$ be defined by $R = \{(x, y) : x \text{ and } y \text{ are students of the same sex}\}$. Check if R is an equivalence relation. OR (III) A function $f : B \rightarrow G$ be defined by $f = \{(b_1, g_1), (b_2, g_2), (b_3, g_1)\}$. Check if f is bijective. Justify your answer.
41.	A class-room teacher is keen to assess the learning of her students the concept of “relations” taught to them. She writes the following five relations each defined on the set $A = \{1, 2, 3\}$: $R_1 = \{(2, 3), (3, 2)\}$ $R_2 = \{(1, 2), (1, 3), (3, 2)\}$ $R_3 = \{(1, 2), (2, 1), (1, 1)\}$ $R_4 = \{(1, 1), (1, 2), (3, 3), (2, 2)\}$ $R_5 = \{(1, 1), (1, 2), (3, 3), (2, 2), (2, 1), (2, 3), (3, 2)\}$ The students are asked to answer the following questions about the above relations : (i) Identify the relation which is reflexive, transitive but not symmetric. (ii) Identify the relation which is reflexive and symmetric but not transitive. (iii) (a) Identify the relations which are symmetric but neither reflexive nor transitive. OR (iii) (b) What pairs should be added to the relation R_2 to make it an equivalence relation ?

- 42.** A school is organizing a debate competition with participants as speakers $S = \{S_1, S_2, S_3, S_4\}$ and these are judged by judges $J = \{J_1, J_2, J_3\}$. Each speaker can be assigned one judge. Let R be a relation from set S to J defined as $R = \{(x, y) : \text{speaker } x \text{ is judged by judge } y, x \in S, y \in J\}$.



Based on the above, answer the following :

- (i) How many relations can be there from S to J ? 1
- (ii) A student identifies a function from S to J as $f = \{(S_1, J_1), (S_2, J_2), (S_3, J_2), (S_4, J_3)\}$ Check if it is bijective. 1
- (iii) (a) How many one-one functions can be there from set S to set J ? 2

OR

- (iii) (b) Another student considers a relation $R_1 = \{(S_1, S_2), (S_2, S_4)\}$ in set S . Write minimum ordered pairs to be included in R_1 so that R_1 is reflexive but not symmetric. 2

- 43.** Let A be the set of 30 students of class XII in a school. Let $f : A \rightarrow N$, N is a set of natural numbers such that function $f(x) = \text{Roll Number of student } x$. On the basis of the given information, answer the following :

- (i) Is f a bijective function ? 1
- (ii) Give reasons to support your answer to (i). 1

- (iii) (a) Let R be a relation defined by the teacher to plan the seating arrangement of students in pairs, where
 $R = \{(x, y) : x, y \text{ are Roll Numbers of students such that } y = 3x\}$.
List the elements of R . Is the relation R reflexive, symmetric and transitive ? Justify your answer. 2

OR

- (iii) (b) Let R be a relation defined by
 $R = \{(x, y) : x, y \text{ are Roll Numbers of students such that } y = x^3\}$.
List the elements of R . Is R a function ? Justify your answer. 2